Questions

Q1.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \qquad x \neq 0$$

RocketRevise

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0$$

(2)

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ

(5)

(Total for question = 7 marks)

(**Q08 WMA11/01, Oct 2023**)

www.RocketRevise.com

Q2.

A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the

equation

$$H^3 = pt^2 + q$$

where p and q are constants,

(a) find, to 3 significant figures where necessary, the value of p and the value of q.

(4)

Exactly T years after the tree was planted, its height was 5 m.

(b) Find the value of T according to the model, giving your answer to one decimal place.

(2)

RocketRevise

(Total for question = 6 marks)

(**Q02 WMA11/01, Jan 2021**)

Q3.

Solve the simultaneous equations

$$y + 4x + 1 = 0$$

$$y^2 + 5x^2 + 2x = 0$$

(Total for question = 6 marks)

www.RocketRevise.com

(**Q03 WMA11/01, Specimen papers**)

Q4.

Use algebra to solve the simultaneous equations

$$y - 3x = 4$$
$$x^2 + y^2 + 6x - 4y = 4$$

compiled by Ahmed Gul www.RocketRevise.com

You must show all stages of your working.

(Total for question = 7 marks)

(Q04 WMA11/01, Oct 2020)

Q5.

RocketRevise

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

www.RocketRevise.com

(5)

(Total for question = 7 marks)

(Q06 WMA11/01, June 2022)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

Q6.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

(a) write down an equation linking x and y

(1)

Given also that the area of the pitch is 7350 m^2

RocketRevise

(b) write down a second equation linking x and y

(1)

(c) hence find the value of x and the value of y

(4)

(Total for question = 6 marks)

(**Q02 WMA11/01, June 2023**)

Q7.

A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

www.RocketRevise.com

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

(a) find the value of a and the value of b.

(4)

(b) State, according to the model, the height of the tree when it was planted.

www.RocketRevise.com

(1)

(Total for question = 5 marks)

(Q02 WMA11/01, Oct 2019)

Q8.

RocketRevise

The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £V, of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

(a) Find the value of p and the value of q.

(3)

Exactly T years after monitoring began, the share price was £2.50

(b) Find the value of T, according to the model, giving your answer to one decimal place.

(2)

(Total for question = 5 marks)

www.RocketRevise.com

(Q03 WMA11/01, Oct 2022)